

Cardiovascular System

The primary roles of the cardiovascular system are to transport nutrients and remove waste and by-products while assisting with maintaining the environment for all the body's functions. The cardiovascular system plays key roles in the regulation of the body's acid–base system, fluids, and temperature, as well as a variety of other physiological functions. This section describes the anatomy and physiology of the heart and the blood vessels.

Heart

The heart is a muscular organ composed of two interconnected but separate pumps; the right side of the heart pumps blood through the lungs, and the left side pumps blood through the rest of the body. Each pump has two chambers: an **atrium** and a **ventricle** ([figure 1.11](#)). The right and left atria deliver blood into the right and left ventricles. The right and left ventricles supply the main force for moving blood through the pulmonary and peripheral circulations, respectively (41).

Valves

The **tricuspid valve** and **mitral valve** (bicuspid valve) (collectively called **atrioventricular [AV] valves**) prevent the flow of blood from the ventricles back into the atria during ventricular contraction (**systole**). The **aortic valve** and **pulmonary valve** (collectively, the **semilunar valves**) prevent backflow from the aorta and pulmonary arteries into the ventricles during ventricular relaxation (**diastole**). Each valve opens and closes passively; that is, each closes when a backward pressure gradient pushes blood back against it, opening when a forward pressure gradient forces blood in the forward direction (41).

Conduction System

A specialized electrical conduction system ([figure 1.12](#)) controls the mechanical contraction of the heart. The conduction system is composed of

- the **sinoatrial (SA) node**—the intrinsic pacemaker—where rhythmic electrical impulses are normally initiated;
- the internodal pathways that conduct the impulse from the SA node to the atrioventricular node;
- the **atrioventricular (AV) node**, where the impulse is delayed slightly before passing into the ventricles;
- the **atrioventricular (AV) bundle**, which conducts the impulse to the ventricles; and
- the **left bundle branch** and **right bundle branch**, which further divide into the **Purkinje fibers** and conduct impulses to all parts of the ventricles.

The SA node is a small area of specialized muscle tissue located in the upper lateral wall of the right atrium. The fibers of the node are contiguous with the muscle fibers of the atrium, with the result that each electrical impulse that begins in the SA node normally spreads immediately into the atria. The conductive system is organized so that the impulse does not travel into the ventricles too rapidly, allowing time for the atria to contract and empty blood into the ventricles before ventricular contraction begins. It is primarily the AV node and its associated conductive fibers that delay each impulse entering into the ventricles. The AV node is located in the posterior septal wall of the right atrium (41).

The left and right bundle branches lead from the AV bundle into the ventricles. Except for their initial portion, where they penetrate the AV barrier, these conduction fibers have functional characteristics quite opposite those of the AV nodal fibers. They are large and transmit impulses at a much higher velocity than the AV nodal fibers. Because these fibers give way to the Purkinje fibers, which more completely penetrate the ventricles, the impulse travels quickly

throughout the entire ventricular system and causes both ventricles to contract at approximately the same time (41).

The SA node normally controls heart rhythmicity because its discharge rate is considerably greater (60-80 times per minute) than that of either the AV node (40-60 times per minute) or the ventricular fibers (15-40 times per minute). Each time the SA node discharges, its impulse is conducted into the AV node and the ventricular fibers, discharging their excitable membranes. Thus, these potentially self-excitatory tissues are discharged before self-excitation can actually occur.

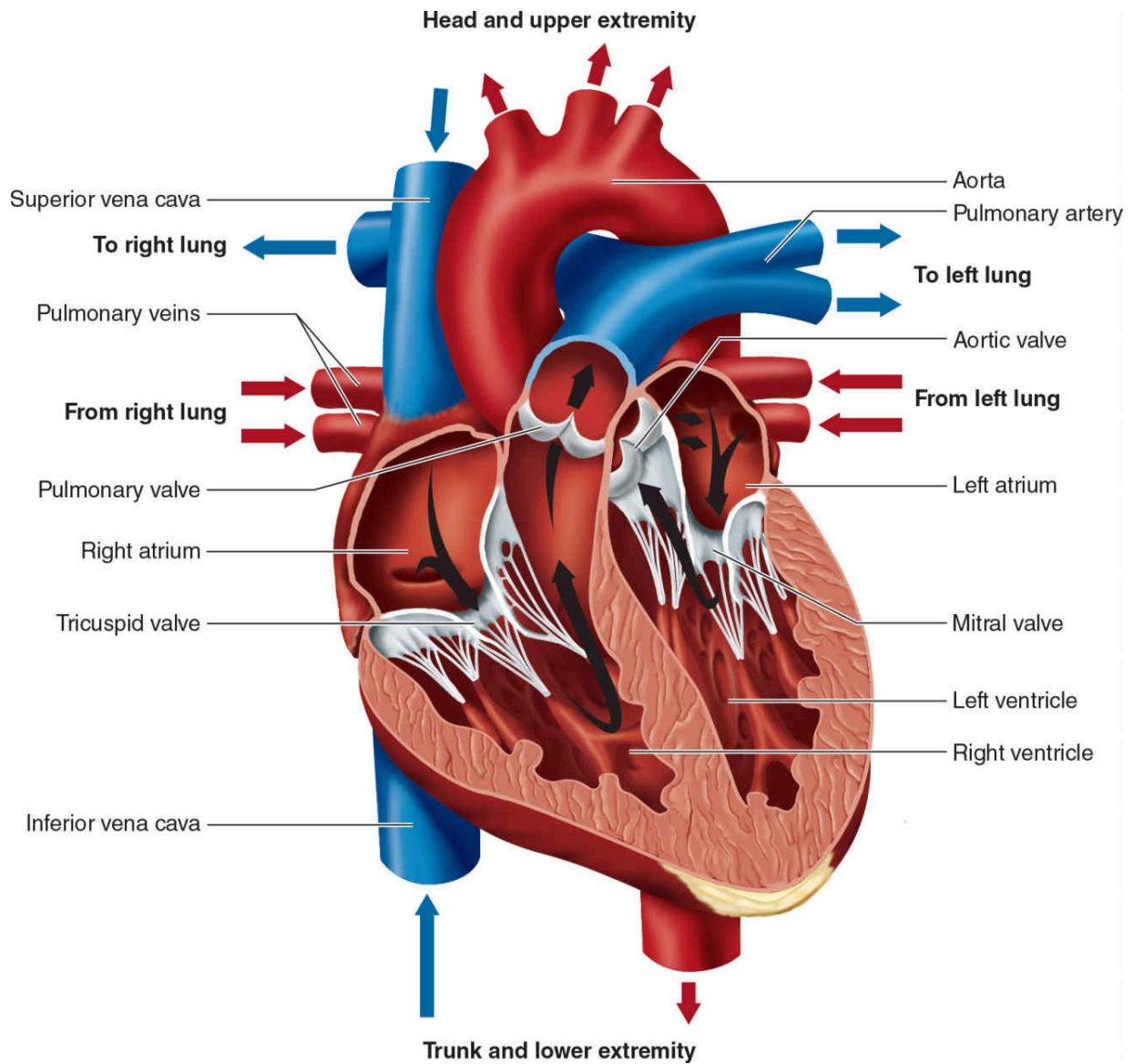


FIGURE 1.11 Structure of the human heart and the course of blood flow through its chambers.

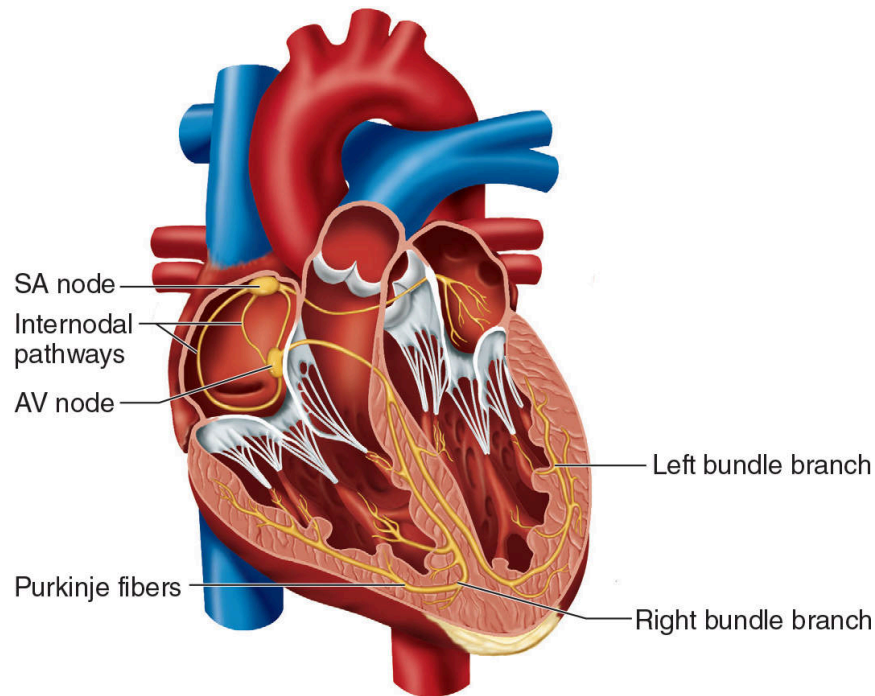


FIGURE 1.12 The electrical conduction system of the heart.

The inherent rhythmicity and conduction properties of the **myocardium** (heart muscle) are influenced by the cardiovascular center of the medulla, which transmits signals to the heart through the **sympathetic** and **parasympathetic nervous systems**, both of which are components of the autonomic nervous system. The atria are supplied with a large number of both sympathetic and parasympathetic neurons, whereas the ventricles receive sympathetic fibers almost exclusively. Stimulation of the sympathetic nerves accelerates depolarization of the SA node (the chronotropic effect), which causes the heart to beat faster. Stimulation of the parasympathetic nervous system slows the rate of SA node discharge, which slows the heart rate. The resting heart rate normally ranges from 60 to 100 beats/min; fewer than 60 beats/min is called **bradycardia**, and more than 100 beats/min is called **tachycardia**.

Electrocardiogram

The electrical activity of the heart can be recorded at the surface of the body; a graphic representation of this activity is called an **electrocardiogram (ECG)**. A normal ECG, seen in [figure 1.13](#), is composed of a **P-wave**, a **QRS complex** (the QRS complex is often three separate waves: a Q-wave, an R-wave, and an S-wave), and a **T-wave**. The P-wave and the QRS complex are recordings of electrical depolarization—that is, the electrical stimulus that leads to mechanical contraction. **Depolarization** is the reversal of the membrane electrical potential, whereby the normally negative potential inside the membrane becomes slightly positive and the outside becomes slightly negative. The P-wave is generated by the changes in the electrical potential of cardiac muscle cells that depolarize the atria and result in atrial contraction. The QRS complex is generated by the electrical potential that depolarizes the ventricles and results in ventricular contraction. In contrast, the T-wave is caused by the electrical potential generated as the ventricles recover from the state of depolarization; this process, called **repolarization**, occurs in ventricular muscle shortly after depolarization. Although atrial repolarization occurs as well, its wave formation usually occurs during the time of ventricular depolarization and is thus masked by the QRS complex (41).

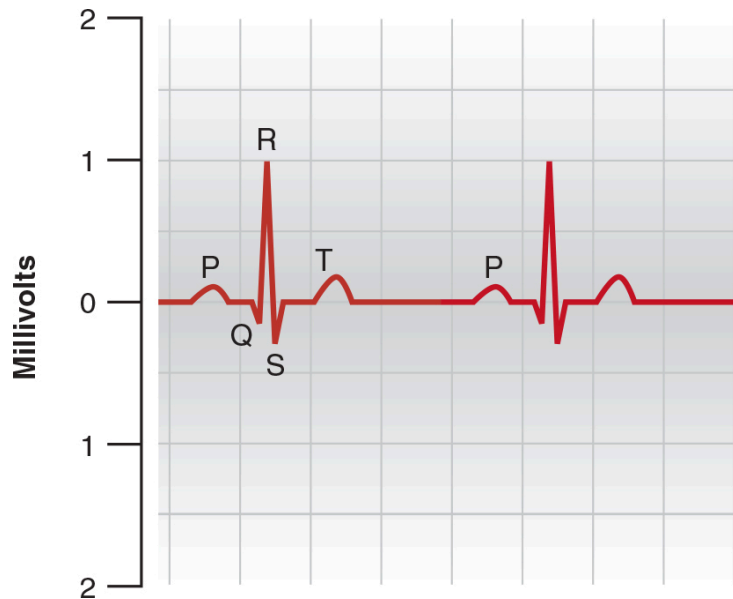


FIGURE 1.13 Normal electrocardiogram.

Blood Vessels

The central and peripheral circulation form a single closed-circuit system with two components: an **arterial system**, which carries blood away from the heart, and a **venous system**, which returns blood toward the heart ([figure 1.14](#)). The blood vessels of each system are identified here.

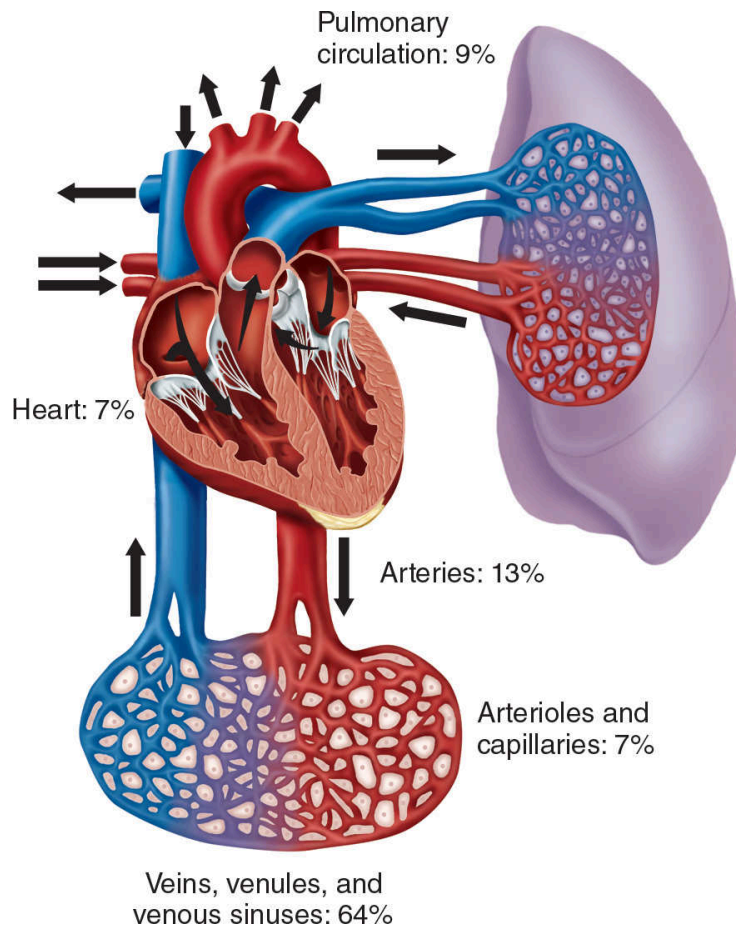


FIGURE 1.14 The arterial (right) and venous (left) components of the circulatory system. The percent values indicate the distribution of blood volume throughout the circulatory system at rest.

Arteries

The function of **arteries** is to rapidly transport blood pumped from the heart. Because blood pumped from the heart is under relatively high pressure, arteries have strong, muscular walls. Small branches of arteries called **arterioles** act as control vessels through which blood enters the capillaries. Arterioles play a major role in the regulation of blood flow to the capillaries. Arterioles have strong, muscular walls that are capable of closing the arteriole completely or allowing it to be dilated many times their size, thus vastly altering blood flow to the capillaries in response to the needs of the tissues (41).

Capillaries

The function of **capillaries** is to facilitate exchange of oxygen, fluid, nutrients, electrolytes, hormones, and other substances between the blood and the interstitial fluid in the various tissues of the body. The capillary walls are very thin and are permeable to these, but not all, substances (41).

Veins

Venules collect blood from the capillaries and gradually converge into the progressively larger **veins**, which transport blood back to the heart. Because the pressure in the venous system is very low, venous walls are thin, although muscular. This allows them to constrict or dilate to a great degree and thereby act as a reservoir for blood, either in small or in large amounts (41). In addition, some veins, such as those in the legs, contain one-way valves that help maintain venous return by preventing retrograde blood flow.

What Is the Skeletal Muscle Pump?

The skeletal muscle pump is the assistance that contracting muscles provide to the circulatory system. The muscle pump works with the venous system, which contains the one-way valves for blood return to the heart. The contracting muscle compresses the veins, but since the blood can flow only in the direction of the valves, it is returned to the heart. This mechanism is one of the reasons why it is important to periodically contract leg muscles or to take walking breaks during prolonged sitting. These activities act to prevent blood pooling in the legs and facilitate venous blood return to the heart.

The cardiovascular system transports nutrients and removes waste products while helping to maintain the environment for all the body's functions. The blood transports oxygen from the lungs to the tissues for use in cellular metabolism. Blood also transports carbon dioxide (mainly as bicarbonate ions) from the

metabolically active tissues to the lungs, where it is removed from the body.

Blood

Two paramount functions of blood are the transport of oxygen from the lungs to the tissues for use in cellular metabolism and the removal of carbon dioxide, the most abundant by-product of metabolism, from the tissues to the lungs. The transport of oxygen is accomplished by **hemoglobin**, the iron–protein molecule carried by the red blood cells. Hemoglobin also has an additional important role as an acid–base buffer, a regulator of hydrogen ion concentration, which is crucial to the rates of chemical reactions in cells. **Red blood cells**, the major component of blood, have other functions as well. For instance, they contain a large quantity of carbonic anhydrase, which catalyzes the reaction between carbon dioxide and water to facilitate carbon dioxide removal.

Respiratory System

The primary function of the respiratory system is the basic exchange of oxygen and carbon dioxide. The anatomy of the human respiratory system is shown in [figure 1.15](#). As air passes through the nose, the nasal cavities perform three distinct functions: warming, humidifying, and purifying the air (41). Air is distributed to the lungs by way of the trachea, bronchi, and bronchioles. The **trachea** is called the *first-generation respiratory passage*, and the right and left main **bronchi** are the second-generation passages; each division thereafter is an additional generation (**bronchioles**). There are approximately 23 generations before the air finally reaches the **alveoli**, where gases are exchanged in respiration (41).

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Exchange of Air

The amount and movement of air and expired gases in and out of the lungs are controlled by expansion and recoil of the lungs. The lungs do not actively expand and recoil themselves but rather are acted on to do so in two ways: by downward and upward movement of the diaphragm to lengthen and shorten the chest cavity and by elevation and depression of the ribs to increase and decrease the back-to-front diameter of the chest cavity (41). Normal, quiet breathing is accomplished almost entirely by movement of the diaphragm. During inspiration, contraction of the diaphragm creates a negative pressure (vacuum) in the chest cavity, and air is drawn into the lungs. During expiration, the diaphragm simply relaxes; the elastic recoil of the lungs, chest wall, and abdominal structures compresses the lungs, and air is expelled. During heavy breathing, the elastic forces alone are not powerful enough to provide the necessary respiratory response. The extra required force is achieved

mainly by contraction of the abdominal muscles, which push the abdomen upward against the bottom of the diaphragm (41).

The second method for expanding the lungs is to raise the rib cage. Because the chest cavity is small and the ribs are slanted downward while in the resting position, elevating the rib cage allows the ribs to project almost directly forward so that the sternum can move forward and away from the spine. The muscles that elevate the rib cage are called *muscles of inspiration* and include the external intercostals, the sternocleidomastoids, the anterior serrati, and the scaleni. The muscles that depress the chest are *muscles of expiration* and include the abdominal muscles (rectus abdominis, external and internal obliques, and transversus abdominis) and the internal intercostals (41).

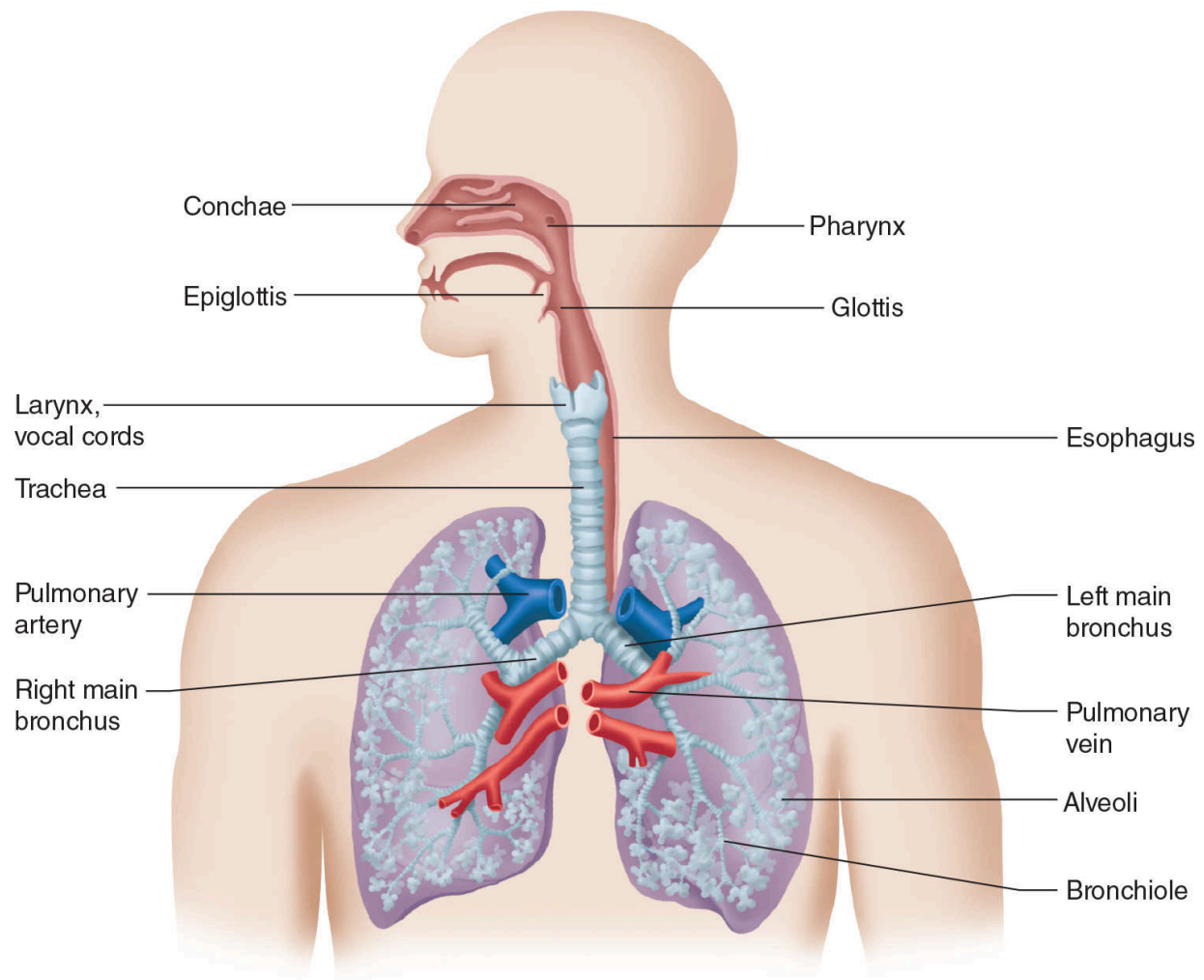


FIGURE 1.15 Gross anatomy of the human respiratory system.

How Important Is It to Train the Muscles of Respiration?

Regular exercise in general is beneficial for maintaining respiratory muscle function. Both endurance exercise, which involves repetitive contraction of breathing muscles, and resistance exercise, which taxes the diaphragm and abdominal muscles especially because of their use for stabilization and for increasing intra-abdominal pressure (Valsalva maneuver), can result in some muscle training adaptations. This can help to preserve some of the pulmonary function with aging.

Pleural pressure is the pressure in the narrow space between the lung pleura and the chest wall **pleura** (membranes enveloping the lungs and lining the chest walls). This pressure is normally slightly negative. Because the lung is an elastic structure, during normal inspiration the expansion of the chest cage can pull on the surface of the lungs and create a more negative pressure, thus enhancing inspiration. During expiration, the events are essentially reversed (41).

Alveolar pressure is the pressure inside the alveoli when the glottis is open, and no air is flowing into or out of the lungs. In fact, in this instance the pressure in all parts of the respiratory tree is the same all the way to the alveoli and is equal to the atmospheric pressure. To cause inward flow of air during inspiration, the pressure in the alveoli must fall to a value slightly below atmospheric pressure. During expiration, alveolar pressure must rise above atmospheric pressure (41).

During normal respiration at rest, only 3% to 5% of the total energy expended by the body is required for pulmonary ventilation. During very heavy exercise, however, the amount of energy required can increase to as much as 8% to 15% of total body energy expenditure, especially if the person has any degree of increased airway resistance, as occurs with exercise-induced asthma. Precautions, including physician evaluation of the athlete, are often recommended, depending on the potential level of impairment.

Exchange of Respiratory Gases

With ventilation, oxygen diffuses from the alveoli into the pulmonary blood, and carbon dioxide diffuses from the blood into the alveoli. The process of **diffusion** is a simple random motion of molecules moving in opposite directions through the alveolar capillary membrane. The energy for diffusion is provided by the kinetic motion of the molecules themselves. Net diffusion of the gas occurs from the region of high concentration to the region of low concentration. The rates of diffusion of the two gases depend on

their concentrations in the capillaries and alveoli and the partial pressure of each gas (41).

At rest, the partial pressure of oxygen in the alveoli is about 60 mmHg greater than that in the pulmonary capillaries. Thus, oxygen diffuses into the pulmonary capillary blood. Similarly, carbon dioxide diffuses in the opposite direction. This process of gas exchange is so rapid as to be thought of as instantaneous (41).

Acute Responses to Aerobic Exercise

A single bout of aerobic exercise places a significant metabolic demand on the body, especially the cardiovascular, respiratory, and muscular systems. Repeated exposure to the acute stress of exercise that occurs with chronic exercise training results in many changes in the function and responses of all body systems. A basic knowledge of the acute effects of aerobic exercise provides the foundation for an understanding of the chronic adaptations that are discussed in [chapter 6](#).

Cardiovascular Responses

The primary functions of the cardiovascular system during aerobic exercise are to deliver oxygen and other nutrients to the working muscles and remove metabolites and waste products. This section describes the cardiovascular mechanisms of these acute responses.

Cardiac Output

Cardiac output is the amount of blood pumped by the heart in liters per minute and is determined by the quantity of blood ejected with each beat (**stroke volume**) and the heart's rate of pumping (**heart rate**):

$$\dot{Q} = \text{Stroke volume} \times \text{Heart rate} \quad (1.1)$$

where $\dot{Q}$ is the cardiac output. Stroke volume is measured in milliliters of blood per beat, and heart rate is measured in beats (contractions) per minute (17).

In the progression from rest to steady-state aerobic exercise, cardiac output initially increases rapidly, then more gradually, and subsequently reaches a plateau. With maximal exercise, cardiac output may increase to four times the resting level of about 5 L/min to a maximum of 20 to 22 L/min (1). Stroke volume begins to increase at the onset of exercise and continues to rise until the individual's oxygen consumption is at approximately 40% to 50% of

maximal oxygen uptake (1). At that point, stroke volume begins to plateau. Sedentary college-aged men have maximal stroke volumes averaging between 100 and 120 mL of blood per beat; maximal stroke volumes for college-aged women are approximately 25% less, due to a smaller average body size as well as a smaller heart muscle (32). The effect of training on the responses to exercise is marked, and we see an increase in maximal stroke volume for college-aged men up to 150 to 160 mL per beat and approximately 100 to 110 for college-aged women (32).

Stroke Volume

Two physiological mechanisms are responsible for the regulation of stroke volume. The first is a result of the **end-diastolic volume**, the volume of blood available to be pumped by the left ventricle at the end of the filling phase, or diastole. The second is due to the action of catecholamines, including epinephrine and norepinephrine, which are hormones of the sympathetic nervous system that produce a more forceful ventricular contraction and greater systolic emptying of the heart.

At the onset of exercise, or even with the anticipation of exercise, sympathetic stimulation increases myocardial contractility and consequently increases stroke volume (14, 27). With prolonged aerobic exercise, the amount of blood returning to the heart (also called **venous return**) is increased due to a combination of venoconstriction (induced via increased sympathetic nervous system activation) (2), the skeletal muscle pump (muscular contractions combine with one-way venous valves to “push” more blood to the heart during exercise (16), and the respiratory pump (increased respiratory frequency and tidal volume) (28). These responses result in pressure alterations on the chambers of the heart and thoracic vena cava, which promote increased venous return (28), and thus end-diastolic volume is significantly increased. With the increased volume, the myocardial fibers become more stretched than at rest, resulting in a more forceful contraction, an increase in force of

systolic ejection, and greater cardiac emptying (17). This principle, called the **Frank-Starling mechanism**, is related to the concept that the force of contraction is a function of the length of the fibers of the muscle wall. This increase in cardiac emptying is characterized by an increase in the **ejection fraction**, or the fraction of the end-diastolic volume ejected from the heart (14, 17).

Heart Rate

Just before and at the beginning of an exercise session, a reflex or anticipatory stimulation of the sympathetic nervous system results in an increase in heart rate. Heart rate increases linearly with increases in intensity during aerobic exercise (14). The rate of increase in heart rate, the actual heart rate response, and the maximal heart rate achieved relate to a variety of individual characteristics of the human system, including fitness and age, in addition to exercise workload.

How Can Athletes Estimate Maximal Heart Rate?

A simple estimate of **maximal heart rate** is to subtract one's age from 220; for example, the estimated maximal heart rate for a 47-year-old person is

$$220 - 47 \text{ (age in years)} = 173 \text{ beats/min.}$$

The variance, or standard deviation, around this estimate is ± 10 to 12 beats/min; thus the actual maximal heart rate for this individual could be expected to fall within the range of 161 to 185 beats/min. A meta-analysis determined that the equation $208 - 0.7 \times \text{age}$ could be used in healthy adults to more accurately predict maximal heart rate (40).

Oxygen Uptake

Oxygen uptake is the amount of oxygen consumed by the body's tissues. The oxygen demand of working muscles increases during an acute bout of aerobic exercise and is directly related to the mass of exercising muscle, metabolic efficiency, and exercise intensity. Aerobic exercise involving larger muscles is likely to be associated

with a higher total oxygen uptake. Increased metabolic efficiency allows for an increase in oxygen uptake, especially at maximal exercise.

Maximal oxygen uptake is the greatest amount of oxygen that can be used at the cellular level for the entire body. Maximal oxygen uptake has been found to correlate well with the degree of aerobic conditioning and is recognized as the most widely accepted measure of cardiorespiratory fitness (14). The capacity to use oxygen is largely related to the ability of the heart and circulatory system to transport oxygen. However, muscular characteristics such as muscle fiber type, muscle fiber mitochondrial density, and capillary density in muscle tissue also affect maximal oxygen uptake. Resting oxygen uptake is estimated at 3.5 mL of oxygen per kilogram of body weight per minute ($\text{mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$) for an average person; this value is defined as 1 **metabolic equivalent (MET)**. Maximal oxygen uptake values in normal, healthy individuals generally range from 25 to 80 $\text{mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$, or 7.1 to 22.9 METs, and depend on a variety of physiological parameters, including age and conditioning level (17).

Oxygen uptake ($\dot{V}\text{O}_2$) can be calculated with the **Fick equation**, which expresses the relationship of cardiac output, oxygen uptake, and arteriovenous oxygen difference:

$$\dot{V}\text{O}_2 = \dot{Q} \times a-\bar{v}\text{O}_2 \text{ difference} \quad (1.2)$$

where $\dot{Q}$ is the cardiac output in milliliters per minute and $a-\bar{v}\text{O}_2$ difference is the **arteriovenous oxygen difference** (the difference in the oxygen content between arterial and venous blood) in milliliters of oxygen per 100 mL of blood. Recalling equation 1.1, we can calculate oxygen uptake as shown in the following example:

$$\dot{V}\text{O}_2 = \text{Heart rate} \times \text{Stroke volume} \times a-\bar{v}\text{O}_2 \text{ difference}$$

$$\dot{V}\text{O}_2 = 72 \text{ beats/min} \times 65 \text{ mL blood/beat} \times 6 \text{ mL O}_2/100 \text{ mL blood} = 281 \text{ mL O}_2/\text{min}$$

To express oxygen uptake in its common unit (i.e., $\text{mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$), we would then divide the result by the person's weight in

kilograms. This is an example for an 80 kilogram (176 pound) athlete:

$$\dot{V}O_2 = 281 \text{ mL O}_2/\text{min} \div 80 \text{ kg}$$

$$\dot{V}O_2 = 3.5 \text{ mL} \cdot \text{kg}^{-1} \cdot \text{min}^{-1}$$

Blood Pressure

Systolic blood pressure estimates the pressure exerted against the arterial walls as blood is forcefully ejected during ventricular contraction (systole) and, when combined with heart rate, can be used to describe the myocardial oxygen consumption (work) of the heart. This estimate of the work of the heart is obtained according to the following equation and is referred to as the **rate–pressure product**, or **double product**:

$$\text{Rate–pressure product} = \text{Heart rate} \times \text{Systolic blood pressure} \quad (1.3)$$

Diastolic blood pressure is used to estimate the pressure exerted against the arterial walls when no blood is being forcefully ejected through the vessels (diastole). Diastolic blood pressure provides an indication of peripheral resistance and can decrease with aerobic exercise due to vasodilation. In systemic circulation, pressure is highest in the aorta and arteries and rapidly falls off within the venous circulation. Also, because pumping by the heart is pulsatile, resting arterial pressure on average fluctuates between a systolic level of 120 mmHg and a diastolic level of 80 mmHg (approximate values). As the blood flow continues through the systemic circulation, its pressure falls progressively to nearly 0 mmHg (venous pressure) by the time it reaches the termination of the vena cava in the right atrium (17).

The **mean arterial pressure** is the average blood pressure throughout the cardiac cycle (equation 1.4). Mean arterial pressure is not the average of systolic and diastolic pressures, because the arterial pressure usually remains nearer the diastolic level than the systolic level during a greater portion of the cardiac cycle. Thus, the

mean arterial pressure is usually less than the average of the systolic and diastolic pressures.

$$\text{Mean arterial blood pressure} = [(\text{Systolic blood pressure} - \text{Diastolic blood pressure}) \div 3] + \text{Diastolic blood pressure} \quad (1.4)$$

Normal resting blood pressure generally ranges from 110 to 139 mmHg systolic and from 60 to 89 mmHg diastolic. With maximal aerobic exercise, systolic pressure can normally rise to as much as 220 to 260 mmHg, while diastolic pressure remains at the resting level or decreases slightly (17, 27).

Control of Local Circulation

Resistance to blood flow is also increased with increasing viscosity of the blood and the length of the vessel. However, these factors remain relatively constant under most circumstances. Thus, **vasoconstriction** (reduction in vessel diameter) and **vasodilation** (increase in vessel diameter) are the primary mechanisms for regulating regional blood flow.

During aerobic exercise, blood flow to active muscles is considerably increased by the dilation of local arterioles; at the same time, blood flow to other organ systems is reduced by constriction of the arterioles. At rest, 15% to 20% of cardiac output is distributed to skeletal muscle, whereas with vigorous exercise this value may rise to 90% of cardiac output (14, 27).

Acute aerobic exercise results in increased cardiac output, stroke volume, heart rate, oxygen uptake, systolic blood pressure, and blood flow to active muscles and a decrease in diastolic blood pressure.

Respiratory Responses

Aerobic exercise provides for the greatest impact on both oxygen uptake and carbon dioxide production as compared to other types of exercise, such as anaerobic resistance training. Significant increases

in oxygen delivered to the tissue, carbon dioxide returned to the lungs, and **minute ventilation** (the volume of air breathed per minute) provide for appropriate levels of alveolar gas concentrations during aerobic exercise (27).

With aerobic exercise, increased minute ventilation occurs as a result of increases in the depth of breathing, frequency of breathing, or both. During strenuous exercise, the breathing frequency of healthy young adults usually increases from 12 to 15 breaths per minute at rest to 35 to 45 breaths per minute, while **tidal volume** (TV), the amount of air inhaled and exhaled with each breath, increases from resting values of 0.4 to 1 L to as much as 3 L or greater. Consequently, the minute ventilation can increase to 15 to 25 times the resting value, or to values of 90 to 150 L of air per minute (14, 17, 27).

During low- to moderate-intensity aerobic exercise, there is an increase in ventilation directly associated with both increased oxygen uptake and carbon dioxide production. In this instance, the increase in ventilation is primarily due to increased tidal volume. The ratio of minute ventilation to oxygen uptake is termed the **ventilatory equivalent** and ranges between 20 and 25 L of air per L of oxygen consumed. In more intense exercise (generally above 45% to 65% of maximal oxygen uptake in untrained individuals and 70% to 90% in trained athletes), breathing frequency takes on a greater role. At these levels, minute ventilation rises disproportionately to the increases in oxygen uptake and begins to parallel the abrupt rise in blood lactate. At this point, the ventilatory equivalent may increase to 35 or 40 L of air per L of oxygen consumed with this high-intensity activity (14, 27).

With inspiration, air enters the alveoli, the functional unit of the pulmonary system where gas exchange occurs. However, with inspiration, air also occupies areas of the respiratory passages: the nose, mouth, trachea, bronchi, and bronchioles. This area is not functional for gas exchange and is called the **anatomical dead space**. The normal volume of this air space is approximately 150 mL

in young adults and increases with age. Because the respiratory passages stretch with deep breathing, the anatomical dead space increases as tidal volume increases ([figure 1.16](#)). Nevertheless, the increase in tidal volume with deep breathing is proportionately greater than any increase in anatomical dead space. Thus, increasing tidal volume (deeper breathing) provides for more efficient ventilation than increasing frequency of breathing alone (17, 27).

Physiological dead space refers to alveoli in which poor blood flow, poor ventilation, or other problems with the alveolar surface impair gas exchange. The physiological dead space in the lungs of healthy people is usually negligible because all or nearly all alveoli are functional. Certain types of lung disease such as chronic obstructive lung disease or pneumonia can significantly reduce alveolar function, increasing physiological dead space by as much as 10 times the volume of anatomical dead space (17, 27).

During aerobic exercise, large amounts of oxygen diffuse from the capillaries into the tissues; increased levels of carbon dioxide move from the blood into the alveoli; and minute ventilation increases to maintain appropriate alveolar concentrations of these gases.

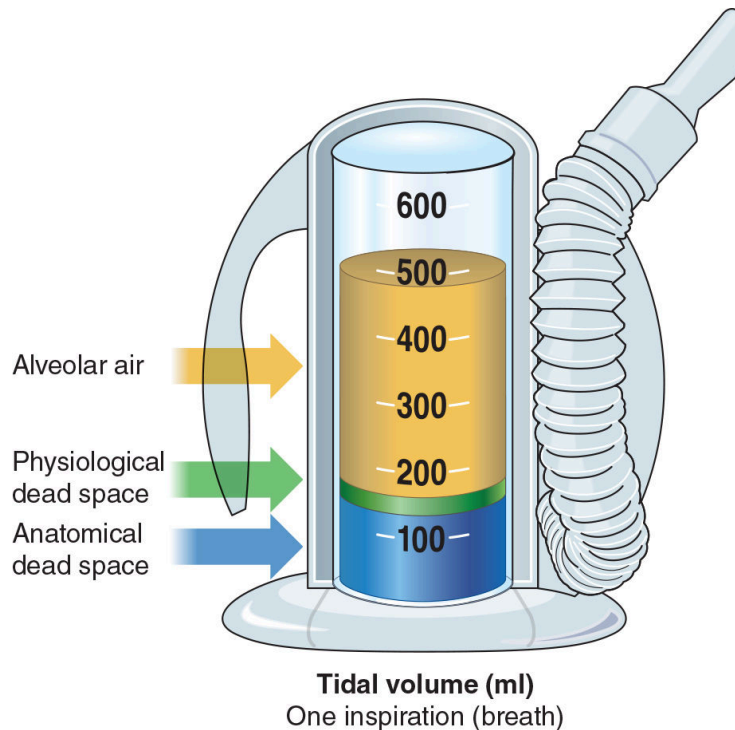


FIGURE 1.16 Distribution of tidal volume in a healthy athlete at rest. The tidal volume comprises about 350 mL of room air that mixes with alveolar air, about 150 mL of air in the larger passages (anatomical dead space), and a small portion of air distributed to either poorly ventilated or incompletely filled alveoli (physiological dead space).

Gas Responses

Diffusion is the movement of oxygen and carbon dioxide across a cell membrane and is a function of the concentration of each gas and the resulting partial pressure exerted by the molecular motion of each gas. Diffusion results from the movement of gas from high concentration to low concentration. At the tissue level, where oxygen is used in metabolism and carbon dioxide is produced, the partial pressures of these gases in some instances differ considerably from those in arterial blood ([figure 1.17](#)). At rest, the partial pressure of oxygen in the interstitial fluid (fluid between muscle fibers) rapidly drops from 100 mmHg in arterial blood to as low as 40 mmHg, while the partial pressure of carbon dioxide is elevated above that of arterial blood to about 46 mmHg. During high-intensity aerobic exercise, the partial pressures of these gases are approximately 3 mmHg for oxygen and 90 mmHg for carbon dioxide. Consequently,

these pressure gradients cause the movement of gases across cell membranes. In addition, the diffusing capacities of oxygen and, in particular, carbon dioxide increase dramatically with exercise, which facilitates their exchange (14, 17, 27).

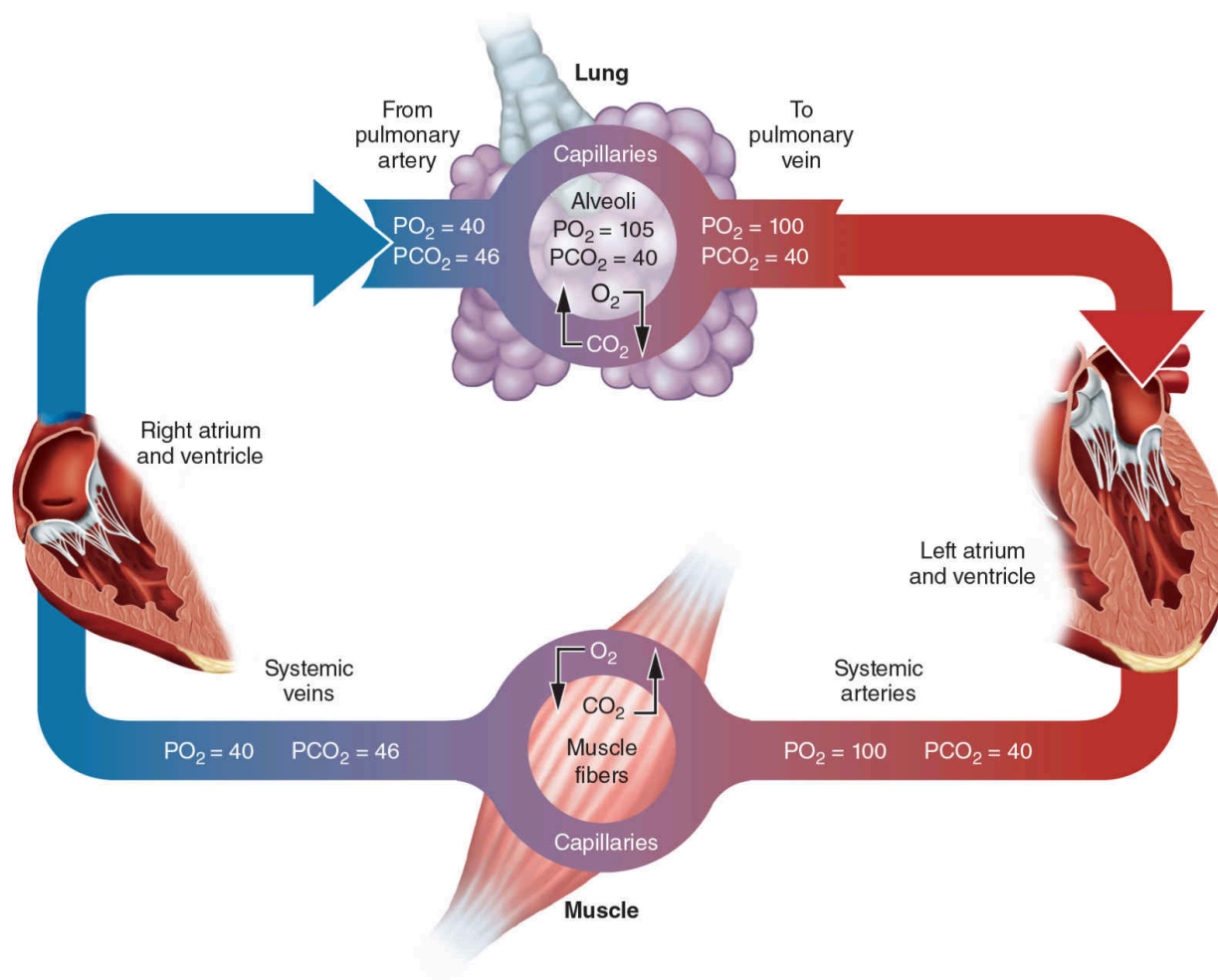


FIGURE 1.17 Pressure gradients for gas transfer in the body at rest. The pressures of oxygen (PO_2) and carbon dioxide (PCO_2) in alveolar air, venous and arterial blood, and muscle tissue are shown.

Blood Transport of Gases and Metabolic By-Products

Oxygen is carried in blood either dissolved in the plasma or combined with hemoglobin. Because oxygen is not readily soluble in fluids, only about 3 mL of oxygen can be carried per liter of plasma.

Nonetheless, this limited amount of oxygen transported in plasma contributes to the partial pressure of oxygen in blood and other body fluids, thus playing a role in the mechanisms that regulate breathing and in the diffusion of oxygen into alveolar blood and the cells of body tissues (17, 27).

Given the limited capacity of plasma to carry oxygen, the majority of oxygen in blood is carried by the hemoglobin. Men have about 15 to 16 g of hemoglobin per 100 mL of blood, and women have about 14 g of hemoglobin per 100 mL of blood. One gram of hemoglobin can carry 1.34 mL of oxygen; thus, the oxygen-carrying capacity of 100 mL of blood is about 20 mL of oxygen in men and a little less in women (27).

The way in which carbon dioxide is removed from the system has some similarities to oxygen transport, but the vast amount of carbon dioxide is removed by a more complex process. After carbon dioxide is formed in the cell, it easily diffuses across the cell membranes and is subsequently transported to the lungs. As with oxygen, only a limited quantity of carbon dioxide—about 5% of that produced during metabolism—is carried in the plasma; similar to the situation with oxygen, this limited amount of carbon dioxide contributes to establishing the partial pressure of carbon dioxide in the blood. Some carbon dioxide is also transported via hemoglobin, but this amount is limited (27).

The greatest amount of carbon dioxide removal (approximately 70%) is from the combination with water and delivery to the lungs in the form of bicarbonate (HCO_3^-). The initial step in this reversible reaction is the combination of carbon dioxide in solution with water in the red blood cells to form carbonic acid. The reaction would normally be quite slow except for the impact of the enzyme carbonic anhydrase, which significantly speeds up this process. Once carbonic acid is formed, it is broken down to hydrogen ions and bicarbonate ions. Because hemoglobin is a significant acid–base buffer, hydrogen ions combine with hemoglobin. This process helps to maintain the pH of the blood. Bicarbonate ions diffuse from the red blood cells to

the plasma, while chloride ions diffuse into the blood cells to replace them (17, 27).

Cardiovascular and Respiratory Responses to Anaerobic Exercise

Acute bouts of anaerobic exercise and long-term anaerobic training both have a significant impact on cardiovascular and respiratory function. This is reflected in both anaerobic athletes (12) and sedentary individuals (22), where enhanced cardiac function and dimensions are apparent. Heavy-load resistance training, resistance training with more repetitions of light loads and less rest, and conventional aerobic endurance training are all beneficial to the cardiovascular system but in different ways (13). Improved ability of the heart, lungs, and circulatory system to function under conditions of high pressure and force production can prepare the athlete's body for the extreme demands of sporting competition.

Acute Cardiovascular Responses to Anaerobic Exercise

An acute bout of anaerobic exercise significantly increases cardiovascular responses. Heart rate, stroke volume, cardiac output, and blood pressure all increase significantly during resistance exercise. Peak blood pressures of 320/250 mmHg and a heart rate of 170 beats/min have been reported during a high-intensity (i.e., 95% of 1RM) leg press exercise (26). Generally, the blood pressure response increases nonlinearly with the magnitude of active muscle mass and is higher during the concentric phase of each repetition than during the eccentric phase, especially at the "sticking point" of an exercise. Although large elevations in blood pressure have been reported during exercise bouts themselves, there are limited data to indicate that resistance training has any negative effects on resting blood pressure (6). In addition, intrathoracic pressure increases, and plasma volume reductions of up to 22% have been reported (34, 36).

Acute anaerobic exercise results in increased cardiac output, stroke volume, heart rate, oxygen uptake, systolic blood pressure, and blood flow to active muscles.

During a set of resistance exercise, stroke volume and cardiac output increase mostly during the eccentric phase of each repetition, especially when the Valsalva technique is used (see [chapter 2](#)) (11). Because the concentric phase of a repetition is much more difficult and elevations in intrathoracic and intra-abdominal pressures are more prominent (via the Valsalva maneuver), limiting venous return and reducing end-diastolic volume, the hemodynamic response of resistance exercise is delayed such that cardiac output increases more during rest periods between sets. This is especially true for an individual's heart rate response; during the first 5 seconds after completion of a set, heart rate is higher than during the set itself (35).

The degree to which blood flow is increased in the working muscles during anaerobic training is dependent on several factors, including the intensity of resistance, the length of time of the effort (i.e., the number of repetitions performed), and the size of the muscle mass activated. When lower resistances are lifted for several repetitions, the responses are relatively similar to those observed during aerobic exercise (15). However, heavy resistance exercise transiently decreases blood flow to the working muscles because of contracted muscle tissue impinging on capillaries and creating a localized occlusion. Muscular contractions greater than 20% of maximal voluntary contraction impede peripheral blood flow within the muscle during a set, but blood flow increases during the subsequent rest period (**reactive hyperemia**) (25). Interestingly, the lack of blood flow (and subsequent increase in metabolites such as hydrogen ions and reduction in pH) during heavy external loading is a potent stimulus for muscle growth (39). Overall, the magnitude of the acute cardiovascular responses depends on the intensity and

volume of exercise, muscle mass involvement, rest period length, and contraction velocity (24, 35).

Ventilatory Response to Anaerobic Exercise

Ventilation rate generally does not limit resistance exercise and is either unaffected or only moderately improved by anaerobic training. With resistance exercise, ventilation is significantly elevated during each set, but the elevation is even greater during the first minute of recovery (35). Ventilation rates of more than 60 L/min have been reported (35), and the rest interval length had a large effect such that short rest intervals (30 s to 1 min) produced the most substantial elevation. Training adaptations include increased tidal volume and breathing frequency with maximal exercise. With submaximal activity, however, breathing frequency is often reduced while tidal volume is increased. It appears that such ventilatory adaptations result from local, neural, or chemical adaptations in the specific muscles trained through exercise (3). Additionally, improved ventilation efficiency, as characterized by reduced ventilatory equivalent for oxygen (the ratio of air ventilated to oxygen used by the tissues, $\dot{V}_E/\dot{V}O_2$), is observed in trained versus untrained individuals (3).

Conclusion

Knowledge of musculoskeletal, neuromuscular, cardiovascular, and respiratory anatomy and physiology is important for the strength and conditioning professional to better understand the scientific basis for conditioning. This includes knowledge of the function of the macrostructure and microstructure of the skeleton and muscle fibers, muscle fiber types, and interactions between tendon and muscle and between the motor unit and its activation, as well as the interactions of the heart, vascular system, lungs, and respiratory system. This information is necessary for developing training strategies that will meet the specific needs of the athlete.

KEY TERMS

- A-band
- acetylcholine
- actin
- action potential
- all-or-none principle
- alveolar pressure
- alveoli
- anatomical dead space
- aortic valve
- appendicular skeleton
- arterial system
- arteriole
- artery
- arteriovenous oxygen difference
- atrioventricular (AV) bundle
- atrioventricular (AV) node
- atrioventricular (AV) valves
- atrium
- axial skeleton

biaxial joints
bone periosteum
bradycardia
bronchi
bronchiole
capillary
cardiac output
cartilaginous joints
crossbridge
crossbridge cycling
depolarization
diastole
diastolic blood pressure
diffusion
distal
double product
ejection fraction
electrocardiogram (ECG)
end-diastolic volume
endomysium
epimysium
extrafusal fibers
fasciculi
fast-twitch fiber
fibrous joints
Fick equation
Frank-Starling mechanism
fused tetanic contraction
Golgi tendon organ (GTO)
heart rate
hemoglobin
hyaline cartilage
H-zone
I-band

inferior
intrafusal fibers
left bundle branch
maximal heart rate
maximal oxygen uptake
mean arterial pressure
metabolic equivalent (MET)
minute ventilation
mitral valve
motor neuron
motor unit
multiaxial joints
muscle fiber
muscle spindle
myocardium
myofiber
myofibril
myofilament
myosin
myosin adenosine triphosphatase
neuromuscular junction
oxygen uptake
parasympathetic nervous system
perimysium
physiological dead space
pleura
pleural pressure
power stroke
proprioceptor
proximal
pulmonary valve
Purkinje fibers
P-wave
QRS complex

rate–pressure product
reactive hyperemia
red blood cell
repolarization
right bundle branch
sarcolemma
sarcomere
sarcoplasm
sarcoplasmic reticulum
semilunar valves
sinoatrial (SA) node
sliding-filament theory
slow-twitch fiber
stroke volume
superior
sympathetic nervous system
synovial fluid
synovial joints
systole
systolic blood pressure
tachycardia
tendon
tidal volume
trachea
tricuspid valve
tropomyosin
troponin
T-wave
twitch
type I fiber
type IIa fiber
type IIx fiber
unfused tetanic contraction
uniaxial joints

vasoconstriction
vasodilation
vein
venous return
venous system
ventilatory equivalent
ventricle
venule
vertebral column
Z-line

STUDY QUESTIONS

1. Based on the sliding-filament theory, which of the following two substances are needed inside of muscle fibers for muscle contraction to occur?
 - a. sodium and fatty acids
 - b. calcium and ATP
 - c. ADP and P_i
 - d. calcium and iron
2. Which of the following is an accurate description of a motor unit?
 - a. a muscle fiber and all of the motor neurons innervating it
 - b. an upper and lower motor neuron
 - c. a motor neuron and all of the muscle fibers it innervates
 - d. a muscle fiber and all of the myofibrils within it
3. Maximal oxygen uptake is most influenced by which of the following?
 - a. the percentage of type I versus type II fibers in an individual's leg muscles
 - b. total blood volume
 - c. age-predicted maximal heart rate
 - d. the ability of the heart and circulatory system to transport oxygen

4. Which of the following respiratory responses occur during an aerobic training bout?
 - a. decreased tidal volume
 - b. decreased physiological dead space
 - c. increased minute ventilation
 - d. decreased motor unit activation of respiratory muscles
5. Which cardiovascular response does not occur during a resistance exercise bout?
 - a. decreased heart rate
 - b. increased stroke volume
 - c. increased cardiac output
 - d. increased blood pressure